Dhaval Patel

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EDUCATION

University of Massachusetts Amherst – Amherst, MA September 2025 – May 2027
Master of Science in Computer Science GPA: 3.76/4.0
Relevant Coursework - Advance Natural Language Processing, Machine Learning, Neural Networks, Software Engineering

Charotar University of Science and Technology – Anand, India October 2021 – May 2025
Bachelor of Technology in Computer Engineering GPA: 9.0/10
Relevant Coursework - Data Structures, Database Management System, Operating System, AI, Image Processing

PROFESSIONAL EXPERIENCE

Cilans System July 2024 - June 2025
AI/ML Engineer Ahmedabad, India

- Architected and deployed scalable backend pipelines for the **GranthAI** document processing platform, integrating multiple third-party APIs (OpenAI, Gemini) to handle high-volume document ingestion and processing.
- Designed and integrated Azure Cosmos DB with scalable indexing and retrieval workflows, supporting multilingual document search at scale with sub-second query response times.
- Reduced end-to-end system latency by 47% by optimizing API call orchestration, implementing efficient caching strategies, and refactoring monolithic workflows into modular components.
- Built RESTful APIs and microservices using FastAPI for document parsing, summarization, and Q&A functionality, ensuring high availability, modular architecture, and seamless downstream integration with frontend services.
- Collaborated in an Agile environment with cross-functional teams, owning feature delivery from design to deployment using Git workflows, peer code reviews, Docker-based containerization, and CI/CD pipelines for reliable, reproducible releases.
- Engineered a proof-of-concept comparing LLM-based OCR with Python OCR libraries, demonstrating 15-20% improvement in accuracy and validating feasibility for production migration.
- Partnered with 2 backend engineers and a client-facing PM to standardize invoice parsing across 8 client formats, increasing extraction accuracy by 21%.

Charotar University of Science and Technology(CHARUSAT) May 2024 - July 2024
Research Intern Anand, India

- Streamlined construction resource estimation accuracy by 24% by extracting dimensional information from floor plan images using OCR and optimizing material requirement calculations for greater precision and efficiency.
- Co-authored 3 papers with a 4-person research team across two labs on applied deep learning, presenting findings on BERT-based cognitive skill evaluation, YOLOv8-driven real-time activity detection, and novel approaches to improving model accuracy, efficiency, and real-world applicability across educational and sports analytics domains.

TECHNICAL SKILLS

Programming Languages: Python, R, C++, HTML, CSS, SQL, Javascript, Node.js
Data Science and Machine Learning: Deep Learning, NLP, LLMs, RAG, Agentic Systems (LangGraph), Prompt Engineering, Model Deployment, Time Series Analysis
Tools & Technologies: Git, Docker, Amazon Web Services (EC2, S3), Microsoft Azure, Claude Code
Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, LangChain, FastAPI, OpenCV, Pandas, NumPy

PROJECTS

DistroTrack — Distributor Management Platform — [Link]

- Developed a platform (Next.js 14, Supabase, TypeScript, Tailwind CSS) digitalizing manual notebook workflows across sales, inventory, center sales, membership tracking, and dashboard analytics with low-stock alerts and volume point tracking.
- Generated period reports for profit, revenue, and inventory with printable summaries, giving distributors accurate figures for business decisions in place of a slow, error-prone paper process.

LedgerLens — Agentic personal finance analyst [Link]

- Built a multi-node LangGraph agent that decomposes natural-language finance questions, routes across SQL / semantic / statistical tools, and self-corrects failed queries — reaching 80.4% execution accuracy across 46 benchmark queries with 100% first-attempt SQL validity, while correctly refusing all 7 deliberately unanswerable ones.
- Built a deterministic verification node rejecting answers whose figures don't trace to a retrieved row: 100% catch rate on 126 injected wrong numbers, 0 false rejections — surfacing hallucinated financial totals instead of passing them silently.

Robust Anomaly Detection under Real-World Image Corruptions — [Link]

- Investigated robustness of industrial anomaly detection models (PaDiM, PatchCore) built on Wide-ResNet-50 against real-world image corruptions such as Gaussian noise, blur, brightness, JPEG across 5 severity levels on the MVTec AD dataset, revealing critical performance gaps where image-level ROC-AUC collapsed to 0.36 below random under brightness corruption for texture categories.
- Designed a corruption-aware data augmentation pipeline that improved model robustness by up to 30% (lifting Gaussian noise AUC from 0.61 to 0.79 and JPEG AUC from 0.55 to 0.78) by training memory banks and Gaussian distributions on mildly corrupted normal samples in PyTorch.

PUBLICATIONS

- **Impact of BERT on Evaluating the Quality of Question Paper Using Bloom's Taxonomy** - ICT4SD-2024 **DOI: 10.1007/978-981-97-8526-1_6**
- **Real-Time Sports Analysis: Integrating YOLOv8 with Enhanced Feature Pyramid Networks for Volleyball and Cricket Activity Detection** - ADCIS-2024 **DOI: 10.1007/978-981-96-4536-7_9**